

Akif Rahim (PhD)

Hydro-Climatologist | Flood and Drought Modeler | Water, Energy, Food and Environment (WEFE) Nexus | Climate Modeling Downscaling and Adaptation

Date of Birth: 4th March 1985.

Education

Dec, 2019 - Jun, 2024

Environmental Modelling (PhD)

Faculty of Environmental Sciences, Czech University of Life Science, Prague, Czech Republic
Phd thesis: „A Rapid Onset Droughts: Unraveling Ecosystem Response to Flash Droughts“.

Sep, 2010 - Aug, 2013

Applied Hydrology (M.phil.)

University of the Punjab, Lahore Pakistan
Thesis: „Optimal Artificial Neural Network Model of Discharge and Sediments load analysis for Indus River at Besham Qila“.

Sep, 2006 - Jul, 2008

Applied Hydrology (M.Sc)

University of the Punjab, Lahore Pakistan
Thesis: „Estimation of flood peaks by using Geomorphological instantaneous unit hydrograph (GIUH) for ungauged catchments of Indus Basin“

Sep, 2009 - Jul, 2010

Diploma in GIS and Remote sensing (PGD)

PIMS, Lahore Pakistan

Sep, 2003 - Jul, 2005

Bachelor of science (B.Sc.)

Summary

As a seasoned Hydro Climate Data Scientist, I specialize in climate extremes, hydrological modeling, ecosystem impacts, and climate change assessment. Proficient in Python and R for hydrological simulation, runoff generation and routing, drought and flood assessment, and regional climate impact modeling using mathematical, numerical, and computational methods. I have hands on experience with top tier hydrological models VIC, HEC-HMS, CLIMADA, and advanced AI/ML applications in hydrology and climate science.

Developed PakDMS, an operational drought monitoring and decision support system for the Pakistan Meteorological Department, and contributed to the Pakistan Water Body Inventory. I also collaborated with Google Pacific Asia to develop an AI driven reservoir operation and flood forecasting system, currently operational with the Punjab Irrigation Department, Pakistan. I am also served as a visiting researcher at the Japan Meteorological Research Institute (MRI), contributing to the development of 5 km × 5 km high resolution climate projections for the Indus Basin. I taught courses in "Exploratory Data Analysis" and "Environmental Data Presentation" to M.Sc. students at the Czech University of Life Sciences Prague.

Work experience

National Researcher | Climate Change and Adaptation

Oct/2024 - Jul/2026

International Water Management Institute (IWMI)
Lahore, Pakistan

- CGIAR Nexus Gain Initiative for INDUS BASIN:

Downscaled high resolution climate projections to assess future water availability in the Indus Basin. Integrated climate projections with hydro economic models to assess future water availability and resource management. Analyzed climate extremes and their projected changes across the Indus Basin to assess population exposure and vulnerability to extreme climate events across the basin.

- Pakistan Drought Monitoring System Pak-DMS (FCDO, UK funded Project):

Designed and developed the drought forecasting module of the Pakistan Drought Monitoring System (Pak-DMS) using the Variable Infiltration Capacity (VIC) hydrological model and AI based surrogate modeling for efficient calibration.

- Assessment of Water Availability (FAO-funded Project):

Assessed water availability in selected watersheds of Balochistan under an FAO funded project. Evaluated surface water availability across ungauged selected river basins. Supported in the mapping of potential aquifer recharge zones.

- Capacity Building of Water Accounting Training (FCDO, UK funded Project):

Designed and delivered a three day climate data and water accounting training for KPK Government stakeholders. Led consultation workshops on Water Resources Assessment in Pakistan for Punjab and KPK water professionals. Strengthened climate resilient planning for agriculture and water security under the FCDO funded WRAP project.

Major: Math,Phy,Stat

University of the Punjab,Lahore
Pakistan

Experties

- ▶ Application of Artificial Intelligence and Machine learning in Hydrological Modeling
- ▶ Application of Remote Sensing and GIS in Hydro-climatology analysis
- ▶ Climate Change Adaptation and Mitigation Policy
- ▶ Sustainability and Environmental Impact studies
- ▶ 1D and 2D River Morphological Analysis
- ▶ Climate Change Impact on the Ecosystem
- ▶ Climate change and Urban water management
- ▶ Water Energy Food Ecosystem Nexus
- ▶ Statistical approaches for hydrological analysis
- ▶ Flood Risk and Vulnerability analysis
- ▶ Design and Implementation of Decision Support System for Water Resource Management
- ▶ Academic and Report writing

Professional-Accomplishments

The following operational tools design and developed

- ▶ Pakistan Drought Monitoring and Decision Support System (PakDMS)
<https://pakdms.iwmi.org>
- ▶ Water Bodies Inventory of Pakistan
<https://pakwmis.iwmi.org/wbip/>

Deputy Director | Meteorology (Hybrid)

Flood Risk Assessment Unit(FRAU)
Government of Punjab Irrigation Department
Lahore,Pakistan

Aug/2021 - Sep/2024

Monitored and forecast river flows across Indus Basin rim stations. Analyzed hydrological data for flood forecasting, reporting, and modeling. Applied SAR based geospatial tools for flood depth estimation and monitored upstream reservoir operations.

PhD Researcher

Hydro Climate Variability Research Group
Czech University of Life Science
Prague, Czech Republic

Dec/2019 - June/2024

Investigate the Flash Drought characteristics across the Central Europe (Project funded by International Grant Agency (IGA) of Czech Republic). Responsible for preparing the Research proposals and analysis. Identify the response of ecosystem to flash droughts in Czech Republic. Design Research methodology for Research proposals. Develop Calibration and validations tools for hydrological modeling. Support to team in the design of data preparation and analysis through Machine and deep learning approaches.

Deputy Director | Meteorology

Flood Risk Assessment Unit(FRAU)
Government of Punjab Irrigation Department
Lahore,Pakistan

Aug/2017 - Nov/2019

Data assimilation of remotely sensed observations into hydrologic, hydrodynamic and agricultural models. Streamflow forecasting in support of reservoir operations using airborne snow observations in Indus western river. Responsible for Monthly, Quarterly and Annual Reports related to flood Risk and flood forecasting. Evaluation of satellite data (GPM,TRRM) for transboundary catchment. Generate the barrage operations regarding to different floods. Assessment of hydrological alteration for River Ravi. Assessment of homogeneous region of rainfall during monsoon. Trend analysis of rain fall and temperature in Punjab. Determine the water availability and droughts for transboundary catchments .

Assistant Director | Flood Routing

Flood Risk Assessment Unit(FRAU)
Government of Punjab Irrigation Department
Lahore,Pakistan

Feb/2016 - Jul/2017

Responsible for Monthly, Quarterly and Annual Reports related to flood Risk and flood forecasting. Construct the semi distributed hydrological model of transboundary catchment of River Chenab. Distributed modeling of adjacent nullah of river Chenab. Flood inundation maps for Indus river. Flood routing model of river Jhelum down stream of Mangla dam. Water distribution model and evaluation efficiency of water distribution plan for Punjab irrigation. SRM models of Chenab and assessment of climate change for water availability. Preparing of flood reports of 2016.

Soil and Water Sanitation Analyst

Estate Accommodator
Islamabad,Pakistan

Mar/2014 - Jan/2016

Soil sampling and testing. Monitoring and evaluation of Water sanitation and supply work. GIS mapping for geological strata of site. Supervise the Construction works of housing Societies in Islamabad. Hydrological modeling and management of water measuring equipment. Data entry and report writing. Preparation of Final Report.

Skills

Surface Water Hydrological Modeling (HEC-RAS, HEC-HMS) 6+ yrs.

R and Python Programming 4+ yrs.

Remote Sensing/ GIS 6+ yrs.

Climate Data Analysis 5+ yrs.

Big Data/ Data science 2+ yrs.

Groundwater Modeling/ Quality Analysis 2+ yrs.

Sustainability Analysis 3+ yrs.

Cloud computing 1+ yrs.

Data Visualization 3+ yrs.

Awards/Scholarships

Letter of Appreciation:

Awarded by Government of the Punjab Irrigation Department, Pakistan

- "In recognition of dedicated services and timely flood forecasts during the 2025 and 2023 events, which significantly contributed to safeguarding lives and critical infrastructure."

Award for Publication Output in Journal Ranked in the 1st Decile by Web of Science (2023).

Awarded by :Czech University of Life Science, Prague, Czech Republic

Title of Paper: Review of GPM IMERG performance: A global perspective"

Special Grant for PhD Research. Awarded by :Czech University of Life Science, Prague, Czech Republic

Duration: 3 years(2019-2022)
Amount: 17,000 Euros

Junior Hydrologist

Rehabilitation of Jinnah Barrage on Indus River
MottMacdonalds of Pakistan(MMP)
Lahore, Pakistan

Implement and review the frame work Real time flow monitoring system (RTFM). Collect the hydraulic data of Jinnah barrage and field survey. Collaborate and conducted meeting with the Punjab irrigation department regarding project progress

Lecturer

Department of Physics
Punjab Group of Colleges
Lahore, Pakistan

Design the study plans of Physics. supervise the examination duties and manage the duty roster of teachers during exams. Organised and managed the recreation trips for students.

Trainings and Workshops

- 03 Aug - 07 Aug-2026: Training of Trainer (TOT) Program on CLIMADA for Improving Climate Governance in Pakistan
GIZ and United Nation University (UNU), Islamabad, Pakistan
- 07 July - 03 Aug-2025: Evaluation of Non-Hydro Dynamics Downscaling of Climate Projection over the Indus Basin
Meteorological Research Institute (MRI), Tasukuba, Japan
- 01 Nov - 31 Dec-2023: Resilience of Ecosystem to Flash Drought over the Central Europe.
Indian Institute of Technology (IIT) Dhanbad, India
- 16 Jan - 03 Mar-2023: Non-Hydro Dynamics Downscaling of Climate Projection
Meteorological Research Institute (MRI), Tasukuba, Japan
- 18-30, Mar-2019: Rapid Assessment of Stream Data for Flood and Drought
Department of Civil Engineering, UET Lahore Pakistan
- 13 - 15, Dec-2018: Water Governance International Water Law and Multi-Stakeholder Global Water Partnership (GWP): Yunnan University, Kunming, China
- 13 - 19, May-2018: Aphrodite -2 Extreme Precipitation analysis for grided data workshop
Hirosaki University, Amori, Japan
- 21 - 23, May-2017: Summer School on Water informatics : Science and System
Center for Water informatics and Technology (WIT) LUMS, Lahore, Pakistan
- 18 - 29, Jul-2016: Remote Sensing and GIS Application on Flood Management
UNESCO short course, NUST institute of Civil Engineering Islamabad Pakistan

Conference presentations

- May-2026: Climate Driven Hydrological Intensification and Its Implications for Water Availability ,28th EGU General Assembly, Vienna, Austria
- May-2026: Kilometer Scale Climate Modeling of Extremes: Evaluation of the Non-Hydrostatic Regional Climate Model (NHRCM) over Pakistan. ,28th EGU General Assembly, Vienna, Austria
- April-2025: Spatiotemporal Evaluation of Vertical Dynamics Propagation of Flash Drought and Driving Mechanisms in the Indus Basin in South Asia (1970–2023). ,27th

Project of "Response of Ecosystem to Flash Drought Across the Central Europe".

- ▶ Awarded by :Internal Grant Agency (Project no:2021B0020), Czech University of Life Sciences Prague, Czech Republic
Duration: 1 year(2020-2021)
Amount: 6,000 Euros

Project of "Non hydrostatic dynamic downscale climate projection for Indus Basin".

- ▶ Awarded by :Ministry of Education, Culture, Sports, Science and Technology (MEXT), Japan
Duration: 2 months (2023)
Amount: One Million Jp Yens
Young Researcher Award (Travel and fee) ".
Awarded by :European Geosciences Union Vienna, Austria
Duration: 1 months (2023)
Amount: 200 Euros

Best Paper Presentation Award on " Leveraging of Remote sensing and AI- Algorithm for Reservoir Operation".

- ▶ Awarded by : Center of RS and Climate Change, University of the Punjab

Project on "Generation of APHRODITE High Resolution Grided data for Pakistan".

- ▶ Awarded by : University of the Hirosaki, Japan
Duration: 1 months (2018)
Amount: 0.5 Million Jp Yens

Language

English

B1

Contact

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EGU General Assembly,Vienna, Austria

■ April-2024: Flash Drought Impacts on Ecosystems: Insights on Resistance and Resilience ,26th EGU General Assembly,Vienna, Austria

■ April-2023: Space and Time Characteristics of Flash Drought Over the Central Europe , 25th EGU General Assembly,Vienna, Austria

■ May-2022: Investigate the Characteristics of Flash Drought Across the Europe , XIth Scientific Assembly of the International Association of Hydrological Sciences (IAHS 2022) Montpellier, France

■ April,2021: Flash Drought Research: Growth, Challenges and Future Perspectives ,23rd EGU General Assembly,Vienna, Austria

■ April-2021: Quantification of water sustainability in upper Indus river basin through the concept of resilience, reliability and vulnerability (RRV) 23rd EGU General Assembly,Vienna, Austria

■ Sep-2020: Global Research Trends in Flash Drought: A Bibliometric Analysis during 2000 to 2019, Conference of Hydrology, GIS and the environment 2020,Prague,Czech Republic

■ Dec-2018: The Impact of Climate change and Indus Water treaty on the Downstream Catchments ,Yunnan University, Kunming, China organized by Global Water Partnership (GWP).
<https://www.gwp.org/en/GWP-China/about-gwp-china/news-list/2018/sharing-water-training-in-kunming>

■ Sep-2018: The Assessment of Water Sustainability of the Himalayan Watershed in Pakistan under the Umbrella of Climate Change Threat ,The 7th World Sustainability Forum, University of International Business and Economics (UIBE). Beijing China.
<https://wsf-7.sciforum.net/section827>

■ Dec-2017:
Day-1:Assessment of future water Availability under Climate Change Case study of Chenab River.

Day-2:The Impact of Climatic variability on Riverian Ecosystem of Kanshi Basin.

Day-3:The Assessment of Hydro Meteorological Drought in Upper Indus Basin on the basis of SPI and SDI drought Index from 1990 to 2013.

International Science-Policy Conference on Climate Change in Pakistan Organized by Global change impact study center(GICSC Pakistan) and The University of UTAH (US) in Islamabad, Pakistan(<http://sp3c.com.pk/>)

■ Nov-2017: The Statistical Approach to asses the hydrological Alteration in the Mangla Watershed of Pakistan , Climate Change Threats to Agriculture and Food Security (Think Adapt 2017) The University of Agriculture Peshawar,Pakistan

Publications

■ Rahim, Akif, et al (2026). "Characterizing the Spatio and Temporal Variability of Flash Droughts Across Europe." Journal of Water and Climate Change (Under Review).

■ Khurshid, Tahira, Qiongfang Li, Chuanhao Wu, Akif Rahim, Muhammad Shafeeqe, Shanshui Yuan, Zia Ul Hassan, and Junliang Jin. "Spatiotemporal Evaluation of Vertical Dynamics Propagation of Flash Drought and Driving Mechanisms in the Indus Basin in South Asia (1970–2023)." EGU sphere 2025

■ Fatima, Zaheen, and Akif Rahim. "Leveraging Remote Sensing and AI Algorithms for Predicting Inflow and Water Levels in Transboundary Reservoirs on Eastern Reivers of Pakistan." Fusion Journal of Engineering and Sciences (2025).

- Rahim, Akif, et al. "Systematic analysis of the flash drought research: contribution, collaboration, and challenges." *Theoretical and Applied Climatology* (2023): 1-14. (Zone-4/Q2)
- Rahim, Akif, et al. "Early 21st Century Trends of Temperature Extremes over the Northwest Himalayas." *Atmosphere* 14.3 (2023): 454 (Zone-4/Q2)
- Rahim, Akif, et al. "The Perturbation of Mangla Watershed Ecosystem in Pakistan Due to Hydrological Alteration." *Water* 15.4 (2023): 656.(Zone-3/Q2)
- Rahim, Akif, Yannis Markonis, and Blöcher Johanna Ruth. "Space and Time Characteristics of Flash Drought Over the Central Europe". No. EGU23-443. Copernicus Meetings, 2023.
- Pradhan, R.K., Markonis, Y., Godoy, M.R.V., Villalba-Pradas, A., Andreadis, K.M., Nikolopoulos, E.I., Papalexiou, S.M., Rahim, A., Tapiador, F.J. and Hanel, M., 2022. Review of GPM IMERG performance: A global perspective. *Remote Sensing of Environment*, 268, p.112754. (Zone-1/Q1)
- Cárdenas Castillero, G., Kuráž, M. and Rahim, A., 2021. "Review of Global Interest and Developments in the Research on Aquifer Recharge and Climate Change": A Bibliometric Approach. *Water*, 13(21), p.3001. (Zone-3/Q2)
- Rahim, A., Tariq, N., Aziz, F., Yousaf, M. and Khurshid, T., 2021, April. "Quantification of water sustainability in upper Indus river basin through the concept of resilience, reliability and vulnerability (RRV)". In *EGU General Assembly Conference Abstracts* (pp. EGU21-393).
- Rahim, A. and Markonis, Y., 2021, April. "Flash Drought Research: Growth, Challenges and Future Perspectives". In *EGU General Assembly Conference Abstracts* (pp. EGU21-14105).
- Aziz, F., Tariq, N., Rahim, A. and Mahmood, A., 2021, April. "Variability of Temperature Extremes in Northwest Himalayas during Early 21st Century". In *EGU General Assembly Conference Abstracts* (pp. EGU21-394).
- Fatima, E., Rahim, A., Hassan, M., Aziz, F. and Yousaf, M., 2021, April. "The Resiliency of High-Altitude Watershed to Dry Condition Under the Changing Climate". In *EGU General Assembly Conference Abstracts* (pp. EGU21-386).
- Tariq, N., Rahim, A., Aziz, F. and Yousaf, M., 2021, April. "Space and Time Characteristics of Droughts in Kabul River Basin". In *EGU General Assembly Conference Abstracts* (pp. EGU21-392).
- Godoy, M.R.V., Pradhan, R.K., Pratap, S., Rahim, A. and Markonis, Y., 2020. "Multi-source quantification of precipitation in the global water cycle" (No. EGU2020-10920). Copernicus Meetings.
- Akif, Amina, Khalida Khan, and Akif Rahim. "THE PRELIMINARY ASSESSMENT OF HYDROLOGICAL ALTERNATION IN THE MANGLA WATERSHED OF PAKISTAN." *Science International* 28.3 (2016).
- Rahim, A., et al. "The geo statistical approach to assess the groundwater drought by using standardized water level index (SWI) and standardized precipitation index (SPI) in the Peshawar regime of Pakistan." *Sci Int* 27.5 (2015): 4111-4117.
- Rahim, Akif, and Amina Akif. "Optimal artificial neural network modeling of sedimentation yield and runoff in high flow season of Indus river at Besham Qila for Terbela dam." *Int J Sci Res* 4 (2015): 479-483.
- Akif, Rahim, et al. "The spatial and temporal variation in the ground water potential due to urbanization in the Peshawar regime of Pakistan." *Science International (Lahore)* 27.3, Section A (2015): 2225-2233.